

After some hours, you will see the WPA PIN cracked and displayed in the terminal.

WPA PSK: X

X is the password of the targeted wireless network.

- **Social engineering**

There are smarter ways of getting the password directly from the user like the Man in the Middle attack. It involves phishing. In this clever attack, you forcefully disconnect a client from the original WPA-2 network, then trick them into connecting to a fake open network created by you. Once connected to the fake network, send the client a made up login page asking him to enter sensitive information. We cannot create a WPA-2 connection directly instead of an open network to get the password directly as WPA-2 performs a mutual authentication during the 4-way handshake. The client verifies whether the AP is legitimate or not, and knows the password (which is hashed), and the AP verifies that the client is legitimate and also knows the password. We don't have enough information to complete the 4-way handshake.

Drawback of hacking WPA

A WEP network could be hacked by any random guy with a laptop, and a network connection, hence the much more robust WPA/WPA2 was developed and adopted.

As seen above, hacking WPA/WPA2 is a tedious job in most cases. A word list/dictionary attack may take days, and still, there is a chance it might fail. Also, good dictionaries are huge occupying a lot of memory. An exhaustive bruteforce may take years depending on the length of the password. Rainbow tables speed things up, but the memory occupied is disastrously large (can be 100s of GBs).

Conclusion

Hacking into a Wi-Fi network is not as easy as shown in the movies and not remotely as fast, but with the right tools at your disposal, it is possible to hack into Wi-Fi networks around you (especially the ones with a weak password). Hacking into others' networks is not advisable without permission. You could use the above-mentioned techniques to check the strength of your Wi-Fi and how secure your password is. Once again, 'Stay Informed, Stay Safe.' 